

Gaussian Process Multi-fidelity Modeling of Near-Earth Galactic Cosmic Ray Spectra

Ian Silber, Department of Physics

*Mentors: Dr. Lucius Schoenbaum, Dr. Vladimir Florinski,
Department of Space Science*

Introduction

Modern machine learning can be leveraged to combine many data sets together to create powerful new platforms. An example of this is multi-fidelity surrogate modeling. Surrogate models are statistical tools that rely on sampling to provide accurate estimates of computational models or empirical measurements that are otherwise prohibitively expensive. Such a case is the computation or measurement of galactic cosmic ray (GCR) energy spectra based on effects of magnetized 3D plasma transport physics and complex solar processes. Because high-energy particles like GCRs are a safety threat for biological organisms as well as nanoscale technology, efficient and accurate estimates of these spectra at a given time near the Earth is valuable for human industrial development and exploration of the solar system.

Low-Fidelity Data

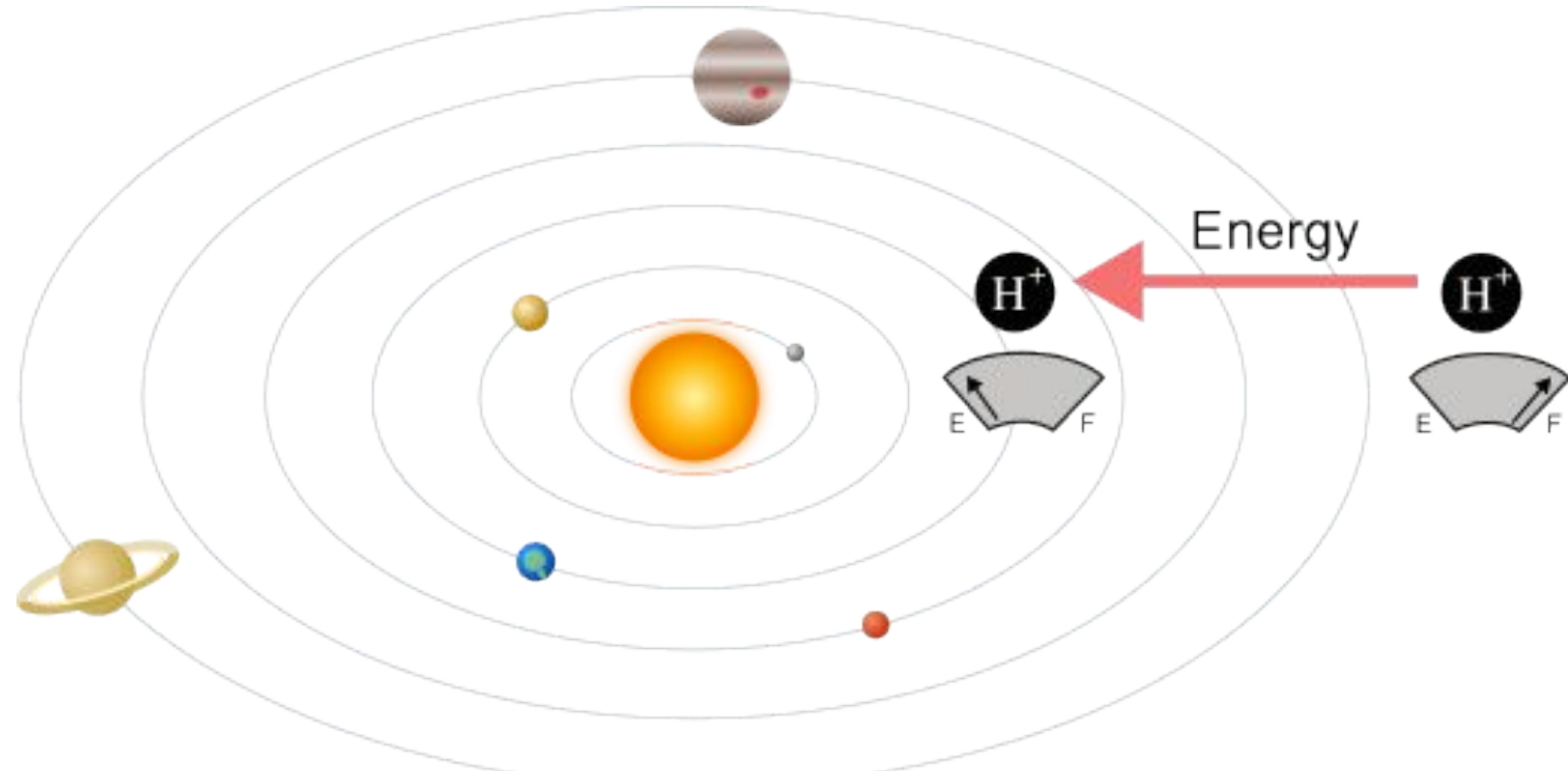


Figure 1. Conceptualization of the Force-Field (FF) approximation for GCR spectra. The spectrum J at radius R for a proton particle species H^+ is a relativistically corrected downward shift of the energy spectra J_{LISM} at the outer edge of the solar system, at the interface with the local interstellar medium (LISM), which has a homogeneous, isotropic spectral profile.

The Force Field (FF) approximation estimates the GCR spectra of particles (for example protons or electrons) based on a coarse average that reduces solar wind transport to a straightforward radially dependent adiabatic energy reduction. It was derived in 1967 from the 3D Parker Transport equation by Gleeson and Axford [1]. It provides accurate estimates of GCR spectra at extremely low computational cost [2], based on the empirical spectrum J_{LISM} at the outermost boundary of the heliosphere. The time-dependent input J_{LISM} is assessed by combining data from space missions (Voyager 1/2, PAMELA, AMS-02) with calibrated GCR modeling [2].

High-Fidelity Data (PAMELA)

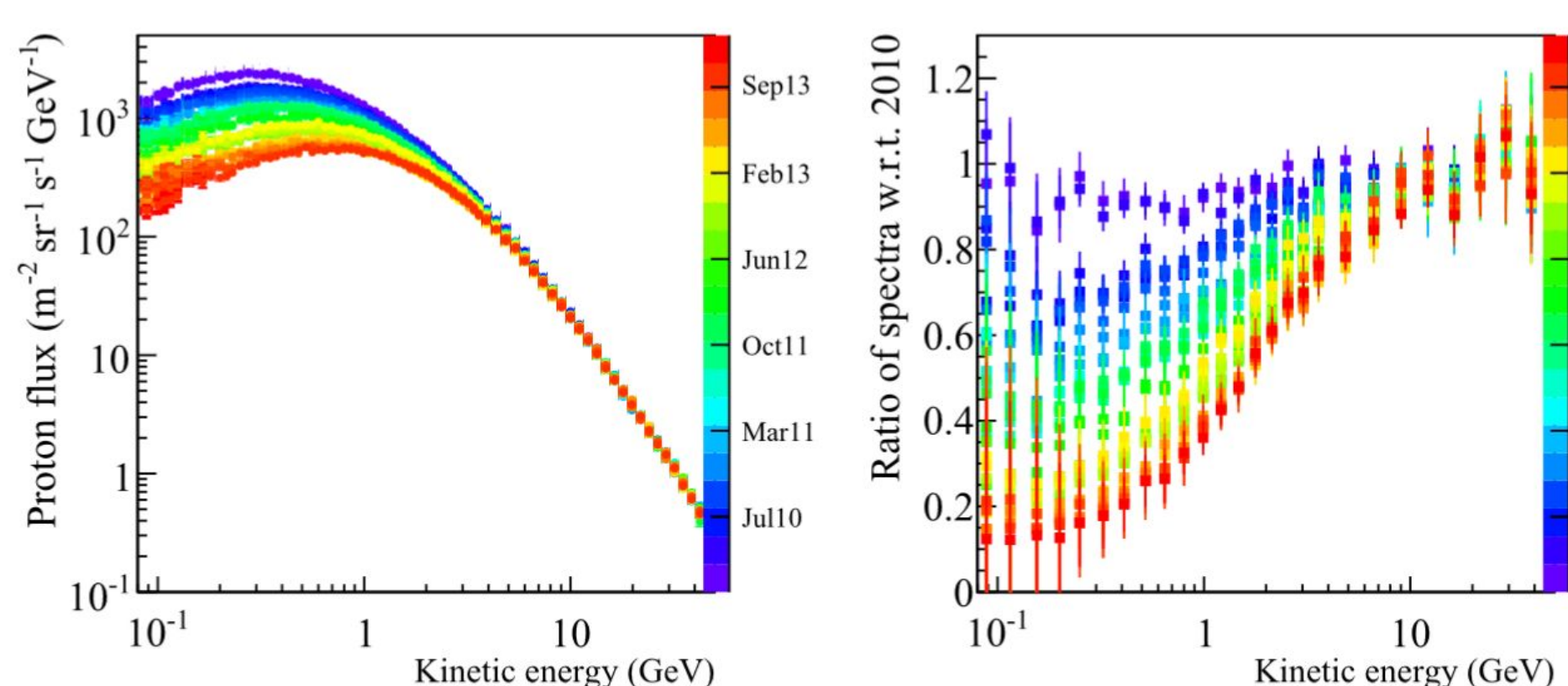


Figure 2. PAMELA data analyzed for years 2010-2013, during the initial solar maximum period of solar cycle 24 (source: [3]).

The PAMELA (Payload for Antimatter Matter Exploration and Light-nuclei Astrophysics) spacecraft was a satellite mission launched on June 15, 2006 by an international collaboration known as WIZARD. During this mission, GCR proton intensity was observed over a wide energy range (0.08–50 GeV) by a single apparatus from a minimum to a maximum period of solar activity for the first time [3].

References

- Gleeson, Axford 1968 *ApJ* **154**
- Puzzoni *et al* 2026 *Adv. Space Res.* **78**(2)
- M. Martucci *et al* 2018 *ApJL* **854** L2
- Forrester *et al* 2007 *Proc. R. Soc. A* **463**(2088): 3251–3269
- Adriani *et al* 2012 *JETPL* **96**(10): 621–627

Acknowledgements

All RCEU projects were sponsored in part by the Alabama Space Grant Consortium, the UAH Offices of the Provost and the Vice President for Research and Economic Development, and the colleges of Engineering; Science; Arts, Humanities, and Social Sciences; Education, Sport, and Human Sciences; Business; and Nursing.

Methods

Gaussian process regression (GPR) is a statistical method for estimating quantities in regions where there is missing information (high uncertainty) by modeling the underlying data source as a Gaussian process. The method leads to a system of equations for weights θ_j

$$\psi^{(i)} = \exp \left(- \sum_{j=1}^k \hat{\theta}_{j(c \text{ or } d)} \|x_j^{(n+1)} - x_j^{(i)}\| \right)$$

Then the prediction is given by

$$\hat{y}_e(x^{(n+1)}) = \hat{\mu} + \mathbf{c}^T \mathbf{C}^{-1} (\mathbf{y} - \mathbf{1}\hat{\mu})$$

Multi-fidelity GPR combines information from data sets of varying fidelity. Low-fidelity data is modeled using a constant scaling factor ρ , plus a Gaussian process $\mathbf{Z}_0$ representing the arithmetic difference between low- and high-fidelity representations. This has the effect of encoding the low-fidelity data as a trend for the GPR model to follow where high-fidelity data gives no information [4].

Results

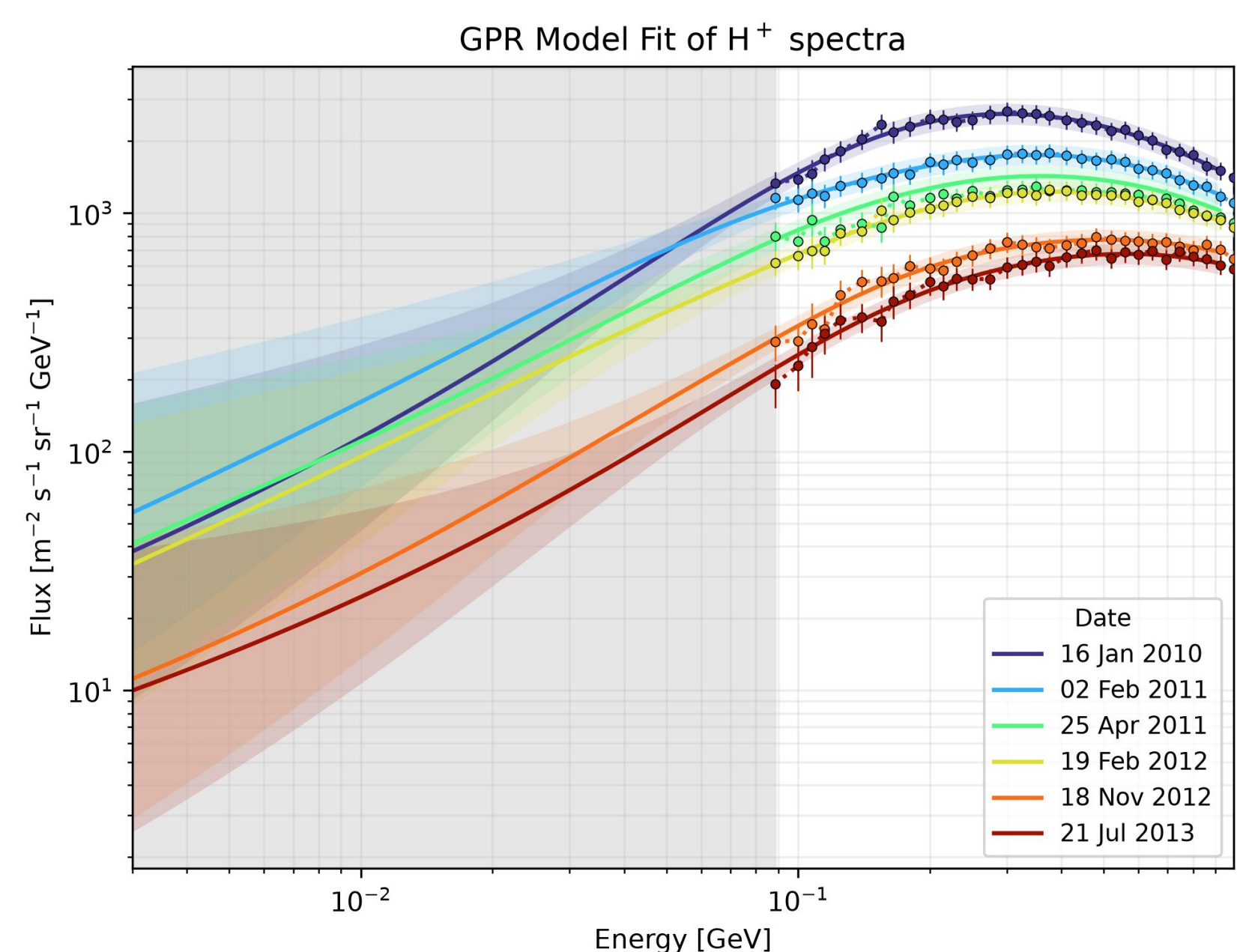


Figure 3. Multi-fidelity GPR model fit, produced using PAMELA data for H^+ (proton) differential energy flux, and the FF approximation, to estimate the spectrum in the low-energy region of interference from high solar activity.

Although the PAMELA apparatus can detect energetic particles in the range ~60 MeV and higher, for GCR spectra in solar cycle 24 it has a lower limit of approximately 300 MeV. Below this energy cannot be resolved due to interference from other particle ensembles, particularly solar energetic particles (SEPs) from high solar activity [3,5]. Using multi-fidelity GPR and the FF approximation, the low-energy region where observations are not available was estimated and uncertainty quantified, see Fig. 3.

In Fig. 4, the force-field approximation is shown along with the GPR model fit and PAMELA data, showing the GPR model's use of the FF model's trend to extrapolate PAMELA mission measurements to the low-energy (sub-100 MeV) region.

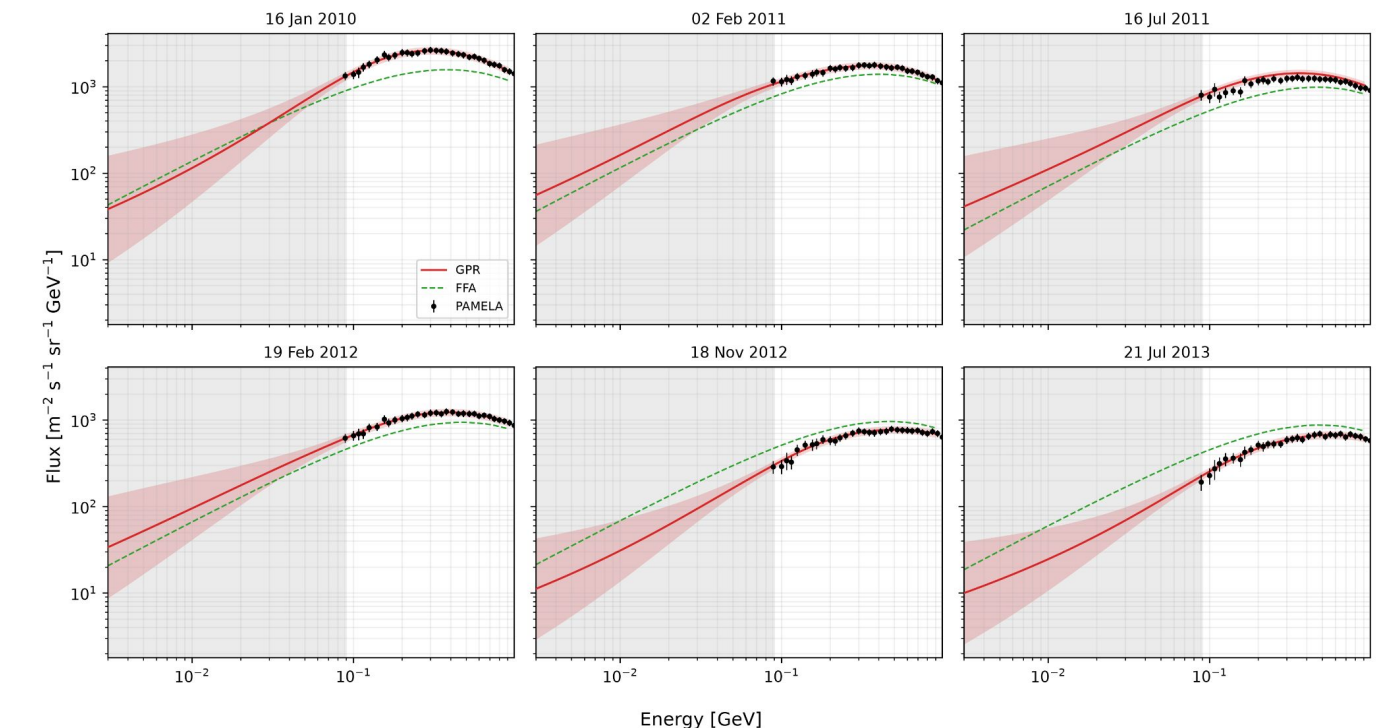


Figure 4. PAMELA data [3] with FF approximation [2], and GPR model curve for six representative date stamps.

Scan to see
source code
on GitHub

